Causal inference and **Topological Data Analysis (TDA)** are two powerful but largely distinct fields that have only recently begun to intersect in meaningful ways. Causal inference (e.g., via potential outcomes, structural causal models/SCMs, or do-calculus) focuses on identifying, estimating, and reasoning about cause-effect relationships from data, often under assumptions like no unmeasured confounding, positivity, consistency, and faithfulness. TDA, rooted in algebraic topology, extracts robust, multi-scale geometric and topological features (e.g., connected components, loops, voids) from complex, high-dimensional, or non-Euclidean data using tools like persistent homology, persistence diagrams, Mapper, and functional summaries (e.g., silhouettes, landscapes).

Their intersection is nascent but promising, particularly for handling complex, structured, or non-Euclidean outcomes where standard scalar or Euclidean summaries (means, variances) miss structural changes. A novel domain like **Topological Causal Data Analysis** could formalize and expand this by integrating TDA's shape-oriented insights with causal frameworks for discovery, estimation, inference, and robustness in high-dimensional or geometric settings.

Brief Overviews of the Fields

Causal Inference: Key frameworks include Pearl's SCMs and do-calculus (for identification and intervention), the potential outcomes framework (Rubin/Neyman for estimation), and causal discovery algorithms (e.g., PC, GES, LiNGAM for time series or general data). Challenges include identifiability (the "causal hierarchy" from associations → interventions → counterfactuals), confounding, selection bias, and scalability to high dimensions or complex data. No assumption-free causal inference is possible in general ("no causes in, no causes out").

Topological Data Analysis (TDA): TDA constructs filtrations (e.g., Vietoris-Rips, Čech, sublevel sets) on point clouds, graphs, images, or time series, computes persistent homology (tracking feature birth/death across scales), and produces persistence diagrams. These are summarized into stable, computable features (e.g., power-weighted silhouettes) for machine learning, clustering, anomaly detection, or visualization (via Mapper). Strengths: robustness to noise, multi-scale view, applicability to non-Euclidean data (graphs, molecules, brain networks). Limitations: computational cost (especially in high dimensions), lack of straightforward statistical inference, and challenges with

interpretability or causality.

Current Intersection and Existing Work

The literature shows limited but growing direct overlap, with two main threads:

1. **Topological Perspectives on Causal Models and Inference**:
   - Ibeling & Icard (2021, NeurIPS): Introduce topologies on spaces of SCMs corresponding to levels of the causal hierarchy. They prove a *topological causal hierarchy theorem*: assumption-free causal inference (e.g., moving from interventional to counterfactual probabilities) is possible only on a meager (topologically "small") set of SCMs. This complements measure-theoretic results and highlights that key assumptions (e.g., faithfulness) are statistically unverifiable in principle. It accommodates infinite variables and clarifies limits of causal learning.
   - This is more "topology on causal model spaces" than TDA on data, providing a meta-level theoretical tool.
2. **TDA for Causal Estimands and Effects on Complex Outcomes**:
   - Kim & Lee (2026, arXiv/ICLR): "Topological Causal Effects." This is the most direct bridge. They define causal estimands via differences in topological structure of *potential outcomes* (e.g., treatment-induced changes in loops/voids in molecular graphs, brain connectivity, or images). Persistence diagrams from filtrations on outcomes are summarized via power-weighted silhouette functions. They provide a doubly robust nonparametric estimator, asymptotic normality/inference, stability bounds, and a test for no topological effect. Applications include macromolecules (treatment changing loop structures), neuroscience, and imaging. This handles non-Euclidean outcomes where mean/variance effects are insufficient.
   - Related: TDA as regularizer in standard ATE pipelines or for robustness under shifts.

3. **Applied/Indirect Combinations**:
   - **Root Cause Analysis (RCA)**: Causal discovery (e.g., via boosting trees) + TDA for clustering high-dimensional data, revealing hidden structures or anomalies with causal explanations. TDA identifies topological irregularities; causal methods explain them. Useful in complex systems (e.g., IT, manufacturing).
   - **Networks and Time Series**: TDA on dependence networks (e.g., brain signals, EEG) or higher-order networks; simulation for statistical inference on topological features in multivariate series. Topology-based causal discovery approaches exist in time series surveys (e.g., using dynamical systems or differential equations).
   - Brain imaging/networks: Persistent homology on connectivity; promises/pitfalls for directed networks and Granger causality links.
   - Other: Anomaly detection, market data, immune response modeling.

Overall, direct papers are few (primarily the two theoretical ones above), but the conceptual fit is strong: TDA excels at "shape" in complex data; causality provides intervention/counterfactual semantics.

Possible Intersections and a Novel "Topological Causal Data Analysis" Domain

Your proposed domain could unify and extend these by treating topological features as first-class citizens in causal pipelines:

- **Causal Discovery with TDA**: Use persistence diagrams or Mapper graphs as inputs/features for discovery algorithms. TDA could detect higher-order structures (cycles, voids) that standard graphs miss, aiding in learning causal mechanisms in dynamical systems or point-cloud data. Combine with topology-based causal methods for time series.
- **Causal Estimation/Inference on Topological Summaries**: Extend Kim & Lee: estimands for treatment effects on full persistence landscapes, bottleneck/Wasserstein distances between diagrams, or Mapper simplicial complexes. Develop identification results, sensitivity analysis, and doubly/multiply robust estimators for functional/topological outcomes.

- **Topological Representations for SCMs or Potential Outcomes**: Embed data into topological spaces for more robust causal modeling (e.g., invariance under deformations).
- **Applications**: Molecular design (causal effects on folding topology), neuroscience (intervention effects on brain topology), climate/spatiotemporal data, genomics, robotics (shape-changing systems), or fairness (topological disparities).
- **Robustness and Generalization**: TDA's stability could improve causal estimators under distribution shift or noise. Topological summaries as outcomes in functional causal inference.

This domain would address gaps in both fields: TDA lacks strong causal semantics and inference; causal inference struggles with complex non-Euclidean data.

Open Problems and Research Questions

Theoretical:

- Identifiability: When can topological features (e.g., Betti numbers, persistence) be causally identified from observational/interventional data? Extensions of the causal hierarchy to topological spaces.
- Statistical inference for TDA in causal settings: Null distributions, confidence regions for diagrams/silhouettes under causal interventions; handling multiplicity across homology degrees.
- Assumptions: Topological analogs of faithfulness, positivity, or SUTVA. Stability of causal conclusions under topological perturbations.
- Scalability and computation: Efficient persistent homology for large causal graphs or high-dim interventions.

Methodological:

- Integrating TDA into discovery algorithms (e.g., constraint-based or score-based with topological scores).
- Handling time-varying or dynamic topologies (persistent homology on evolving filtrations under interventions).

- Multi-modal data: Combining TDA with causal graphs from heterogeneous sources (e.g., images + tabular).
- Counterfactual topologies: Generating or imputing topological structures under "what if" scenarios.

Practical/Open Challenges:

- Interpretability: Linking topological changes (e.g., "birth of a loop") to domain mechanisms.
- Benchmarking: Datasets with ground-truth causal topological effects.
- Software: Unified libraries for topological causal pipelines.
- Limits: When does TDA add value over simpler geometric/ML methods in causal tasks? (See debates on TDA utility.)

Broader: No-free-lunch style results for topological causal learning; ethical considerations in high-stakes applications (e.g., health).

Recommendations for Your Paper

Position "Topological Causal Data Analysis" as a unifying framework bridging these, with the Kim & Lee and Ibeling & Icard works as foundational. Emphasize novelty in systematic integration, new estimands/algorithms, and applications to complex data. Cite TDA surveys (e.g., Carlsson, Chazal) and causal surveys for context. Empirical validation on molecular/neuro data would strengthen it. Future work could focus on open problems above for high impact.

This review is based on key arXiv papers and applications up to 2026; the field is rapidly evolving, so checking recent arXiv searches for "topological causal" or TDA + causality is advised.